Zepto Catalog Health Audit

Inventory, Pricing and Availability Analytics

An End-to-End E-commerce Analytics Project

PostgreSQL (SQL) • Microsoft Excel

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1. Project Overview

Quick-commerce catalogs change constantly and are rarely clean. The same product can be listed under several categories, prices are stored in different units, and stock status changes by the hour. This project audits a real-world product catalog scraped from Zepto, a quick-commerce platform, to find where the catalog loses value and what should be fixed first.

The workflow mirrors a professional analytics pipeline. The raw data is loaded into PostgreSQL untouched, cleaned into two analysis-ready tables, explored, and queried to answer ten structured business questions. The results are then exported to Microsoft Excel, where they power a one-page dashboard and an adjustable restock-priority model.

2. Problem Statement

A category manager at a quick-commerce company needs to understand the health of the product catalog. Product listings appear duplicated, availability varies by category, and discounts look uneven. Leadership needs to know which problems are real, which are data artefacts, and where to act first.

The guiding business question throughout this project is:

> *"Where is the catalog leaking value: through messy data, stock gaps, pricing behaviour, or inventory concentration, and what should be fixed first?"*

3. Dataset

The dataset was sourced from Kaggle and was originally scraped from Zepto's product listings. It contains 3,732 product listings across 9 attributes and 14 categories. Each row is one listing, and the same product can appear several times because it is listed under different categories, which is how real catalog data behaves.

Column	Description
Category	Product category (Fruits & Vegetables, Biscuits, Beverages, and so on)
name	Product name as shown in the app
mrp	Maximum retail price, stored in paise
discountPercent	Discount applied on MRP
availableQuantity	Units available (capped at 6 in this data)
discountedSellingPrice	Final price after discount, stored in paise
weightInGms	Pack weight in grams
outOfStock	Boolean flag for stock availability
quantity	Number of units per pack

Initial profiling showed no NULL values, but it raised an important question: the file had 3,732 rows but only 1,674 distinct product names. This gap pointed to heavy duplication and shaped the whole project.

4. Tools and Technologies

- **PostgreSQL 16** – relational database for storage, cleaning and analysis
- **pgAdmin 4** – database administration, CSV import and result export
- **SQL techniques** – CTEs, window functions (ROW_NUMBER, RANK, LAG, running totals), PERCENTILE_CONT, CORR, FILTER, STRING_AGG and views
- **Microsoft Excel** – structured tables, PivotCharts, Pareto chart, INDEX/MATCH, RANK.EQ, COUNTIFS and the restock model
- **Git and GitHub** – version control and portfolio publication

5. Data Cleaning and Preparation

The cleaning approach was non-destructive. The raw table was never modified; new tables were built from it, so every script can be re-run safely.

1. Loaded the CSV into a raw table (zepto_raw) with prices kept as integer paise, exactly as in the file.
2. Profiled the data: row counts, NULL checks, distinct categories, in-stock versus out-of-stock counts and repeated product names.
3. Logged every problem found in a data_issues table, recording the issue, the number of rows and the action taken.
4. Built zepto_clean (one row per listing): converted paise to rupees, trimmed product names, removed the zero-price row and removed exact duplicate rows with ROW_NUMBER().
5. Calculated the real discount from the two prices and flagged rows where it differed from the stated discount.
6. Built zepto_products (one row per unique product) by grouping listings that match on everything except category, adding listing_count and category_count.
7. Reconciled row counts at each stage: 3,732 raw rows, then 3,729 clean listings, then 1,800 unique products.

Issue found	Rows	Action taken
MRP or selling price is zero	1	Removed from clean table
Weight is zero	4	Kept; excluded from price-per-gram analysis
Product name with leading or trailing spaces	296	Trimmed
Exact duplicate rows (all columns identical)	2	Removed
Stated discount differs from real discount by more than ₹1	408	Kept and flagged (rounding in source, never more than 1 percentage point)
Same product listed under more than one category	1,281	Kept as listings; collapsed to one row in the products table

Table 1: Data quality log (data_issues)

Why two clean tables? zepto_clean holds one row per listing (what a shopper sees on a category page). zepto_products holds one row per unique product (what actually exists). Counting listings when the question is about products inflates every number, so each query uses the table that matches its question.

6. Exploratory Data Analysis – Key Insights

- The 3,729 clean listings represent only 1,800 unique products, an average of 2.07 listings per product.
- 71.2% of products (1,281) appear under more than one category, and one product appears under as many as 7 categories.
- 9 of the 14 categories are made up entirely of cross-listed copies. Biscuits is the cleanest, with no cross-listed products.
- Example: Arden Eggs White is really two products (a 12-egg pack and a 6-egg pack), yet it appears as 10 listings across 5 categories, including Beverages and Munchies. Category labels in this dataset are therefore partly unreliable.
- About 12% of listings are out of stock (453 of 3,732 raw rows).
- Available quantity never exceeds 6 (1,620 rows show exactly 6), so any stock-value figure is an index, not true inventory.
- The 408 “discount mismatches” are rounding: the file stores the discount rounded down to a whole number, and the gap is never more than 1 percentage point.

7. Data Analysis Using SQL

With the clean tables built in PostgreSQL, ten structured business questions were answered. Each query states whether it works on listings or on unique products. Screenshots of the pgAdmin result grids can be pasted under any section below.

7.1 How Inflated Is the Catalog?

Compared the number of listings with the number of unique products, and measured how many products are copied across categories.

SQL technique: *subquery, FILTER clause, JOIN on product_id*

Measure	Result
Total listings	3,729
Unique products	1,800
Average listings per product	2.07
Products in more than one category	1,281 (71.2%)
Most categories for one product	7
Categories that are 100% cross-listed copies	9 of 14

7.2 Which Categories Run Out of Stock Most?

Calculated the out-of-stock rate per category against the overall rate (about 12.2%), and added a label-reliability check so that duplicated categories are not over-interpreted.

SQL technique: *CTE, CROSS JOIN, RANK() window function, CASE*

Category	Out of stock	Label quality
Biscuits	28.6%	Reliable label
Dairy, Bread & Batter	21.7%	Heavily duplicated
Beverages	21.7%	Heavily duplicated
Meats, Fish & Eggs	19.0%	Mixed

Top four categories by out-of-stock rate

7.3 Promoted Products That Are Out of Stock

Found unique products discounted 20% or more that cannot currently be bought. Nine products meet this condition; the four with the deepest discount (50%) are shown below.

SQL technique: *filtering on the unique-products table, multi-column ORDER BY*

Product	Discount
RRO Mozzarella Block Cheese	50%
Chef's Basket Durum Wheat Penne Pasta	50%
Dukes Waffy Chocolate Wafer Rolls	50%
Epigamia Fruit Yogurt Strawberry	50%

7.4 How Deep Are the Discounts?

Grouped unique products into discount bands to see how common deep discounts are, and how many products have no discount at all.

SQL technique: *CASE bands, SUM(COUNT(*)) OVER () for percentage of total*

Discount band	Products	Share
No discount	536	29.8%
1–10%	852	47.3%
11–20%	311	17.3%
21–30%	64	3.6%
Above 30%	37	2.1%

7.5 Do Pricier Products Get Smaller Discounts?

Tested whether discount depth falls as price rises, using price bands and, per category, a correlation between MRP and discount. The common belief that expensive items are discounted less is not supported: the pattern is uneven. Correlations were positive in most categories, and negative only for Fruits & Vegetables (-0.31).

SQL technique: *median with PERCENTILE_CONT, CORR(), HAVING*

Price band (MRP)	Products	Average discount
Under ₹100	849	6.5%
₹100–249	652	8.0%
₹250–499	239	11.0%
₹500–999	47	7.2%
₹1,000 and above	13	14.4%

The top band has only 13 products, so it should be read with care

7.6 Best Deals: Percentage vs Rupees Saved

Ranked in-stock products twice, once by discount percentage and once by rupees actually saved, to show that a percentage ranking hides the deals that save customers the most money.

SQL technique: *two RANK() window functions in one query*

Product	Saved	Discount	Rank by %	Rank by ₹
Borges Extra Light Olive Oil	₹1,201	46%	23	1
Pampers Pants Large	₹700	45%	24	2
Praakritik A2 Ghee	₹145	10%	386	12

7.7 Value for Money Within Each Category

Calculated price per 100g for products of 100g and above, then compared each product with its own category's median, so that salt is never compared with saffron. The three cheapest and three priciest products per category were listed.

SQL technique: *three chained CTEs, PERCENTILE_CONT, ROW_NUMBER() OVER (PARTITION BY category), NULLIF*

Example (Cooking Essentials)	Price per 100g	Versus category median
Aashirvaad Iodised Salt (cheapest)	₹1.90	0.07×
Delight Nuts Dried Blueberries (priciest)	–	11.25×
Category median	₹25.60	1.00×

7.8 Does a Bigger Pack Really Cost Less?

For products sold in several sizes, compared each pack's price per 100g with the next smaller pack. Across 101 size comparisons, about 29% of the time the bigger pack costs more per 100g. The largest case was Godrej No.1 Bathing Soap, where moving from 500g to 600g raised the price per 100g by 47.8%.

SQL technique: *LAG() over a named WINDOW, CASE*

Verdict	Comparisons	Share
Bigger pack cheaper	69	68.3%
Same rate	3	3.0%
Bigger pack costs more	29	28.7%

7.9 Where Is Stock Value Concentrated? (ABC Analysis)

Ranked all 1,583 in-stock unique products by stock value index (selling price × available quantity) and classified them with a running cumulative percentage: Class A up to 70% of value, Class B up to 90%, Class C the rest. Because available quantity is capped at 6, this is a value index rather than true inventory value.

SQL technique: *running total with SUM() OVER (ORDER BY ...), view creation, CASE*

Class	Products	Share of products	Share of value
A	531	33.5%	70.0%
B	428	27.0%	20.0%
C	624	39.4%	10.0%

7.10 Restock Candidates

Listed every out-of-stock unique product with the signals needed for prioritisation: discount percentage, price and listing count. This produced 217 restock candidates. The scoring itself is done in Excel (Section 8.2), where the weights can be changed interactively, which SQL cannot do.

SQL technique: *filtered SELECT feeding an Excel scoring model*

8. Dashboard in Excel

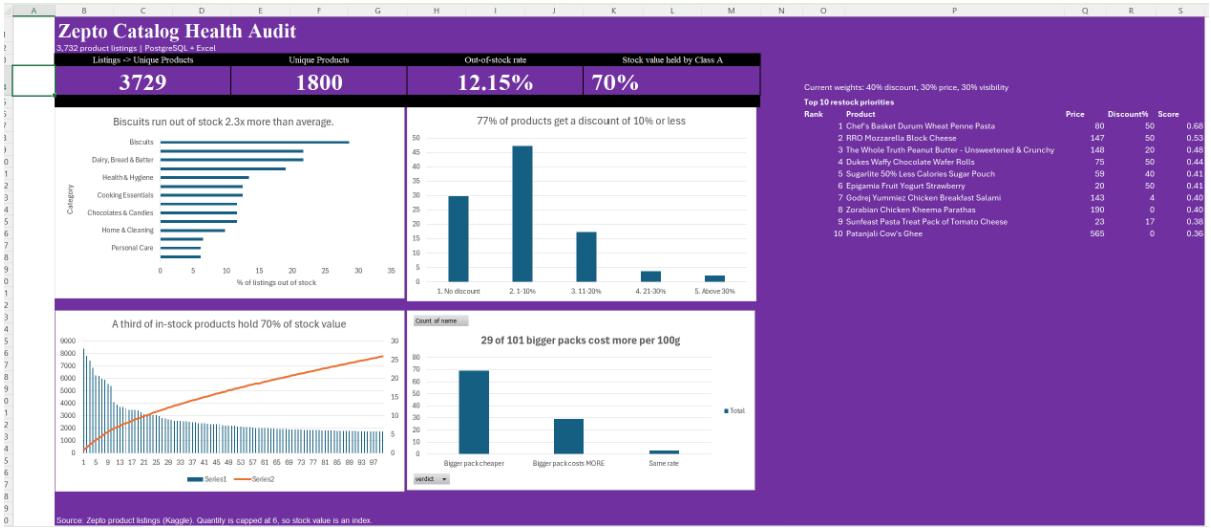
The query results were exported from pgAdmin as CSV files and loaded into Excel as structured tables. They feed a one-page dashboard and a restock-priority model.

8.1 Dashboard

The dashboard fits on one screen and uses a single accent colour with grey, so the finding in each chart stands out. Chart titles state the finding rather than the field name.

Visual	Source	What it shows
Four KPI cards	7.1, 7.2, 7.9	Total listings, unique products, out-of-stock rate, share of value held by Class A
Out-of-stock bar chart	7.2	Availability by category, with Biscuits highlighted
Discount bands column chart	7.4	How discounts are distributed across products
Pareto chart	7.9	Stock value by product with a cumulative percentage line
Pack-size PivotChart	7.8	How often a bigger pack is cheaper, equal or dearer
Top 10 restock table	8.2	Live ranking that updates when weights change

Figure 1: Zepto Catalog Health Audit dashboard (Excel)



8.2 Restock Priority Model

Each of the 217 out-of-stock products receives a priority score from three signals. Each signal is rescaled to a 0–1 range (min-max normalisation) so that discount percentage, price in rupees and listing count can be combined fairly. The weights sit in three input cells; changing them re-ranks the list and updates the dashboard instantly.

Signal	Reasoning	Default weight
Discount depth	How heavily the product is being promoted	40%
Selling price	Revenue earned per sale	30%
Listing count (visibility)	How many places the product appears	30%

The weights are a judgement, not a fact. Products that stay near the top under several different weightings are the most dependable restock picks.

9. Results and Conclusion

9.1 Key Findings

- The catalog is about twice as large as its real product list: 3,729 listings represent 1,800 products, and 71.2% of products are listed under more than one category. Category-level metrics are therefore inflated.
- Biscuits has the clearest availability problem: 28.6% of listings are out of stock, 16.4 percentage points above the overall rate, and its category label is clean.
- Promotions are being wasted: nine products discounted by 20% or more are currently out of stock.
- Discounts are shallow and uneven: about 30% of products have no discount and about 77% have 10% or less. Discount depth does not rise steadily with price.
- Percentage discounts mislead: the largest rupee savings, such as the olive oil at ₹1,201 off, are not the largest percentage discounts.
- “Buy bigger to save” is not always true: in 29 of 101 pack-size comparisons the larger pack costs more per 100g.
- A third of in-stock products (531 Class A products) hold 70% of the stock value index.

9.2 Business Recommendations

- Fix catalog hygiene first: de-duplicate cross-listed products and correct category assignments so that category reporting can be trusted.
- Restock the nine heavily discounted out-of-stock products and the top of the restock model’s ranking before running further promotions.
- Review Biscuits supply to understand why availability is so much lower than the overall rate.
- Audit pack pricing for the 29 cases where larger packs cost more per 100g.
- Protect availability of Class A products, because they carry most of the stock value.

- Feature products with the largest rupee savings in marketing, rather than ranking deals by percentage alone.

9.3 Limitations

- Category labels are partly duplicated, so category-level results are read with caution (section 7.2 includes a label-reliability column).
- Available quantity is capped at 6, so stock value is an index rather than true inventory value.
- There is no cost or sales data, so margins and lost revenue cannot be calculated.
- The data is a single snapshot, so trends over time are not visible.
- The restock weights are a judgement and should be agreed with the business.

10. Author and Contact

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GitHub Repository: https://github.com/Vipparti-Vinay/Zepto-Catalog-health-audit-SQL-Excel

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